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Tips on clean code writing in Python

*“Computers are good at following
instructions, but not at reading your mind”*

Donald Knuth

April 21th. 9pm GMT-5



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Introduction

What is a programming languages?



Introduction

“For decades of using the term programming languages, we thought that they were meant to communicate our ideas to machines so they can run our programs. We were wrong. That's not the truth, but part of the truth. The real meaning of the "language" part of "programming languages" is to communicate our ideas to other developers.”

Source: Clean Code in Python Develop maintainable and efficient code Second Edition - Mariano Anaya



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The meaning of clean code

- There is no sole or strict definition of clean code.
- There is probably no way of formally measuring clean code.
- Clean code is not something a machine or script can recognize but a professionals can decide.



The importance of having clean code

- Maintainability
- Reducing technical debt
- Working effectively with agile development,
- Managing a successful project.



Technical debt

- Refers to the concept of problems in the software as a result of a compromise or a bad decision being made.
- It is a silent problem,

TWO WAYS TO SEE THE TECHNICAL DEBT:

- **From the Present to the past:** what if the problems we are currently facing are the result of previously written bad code?
- **From the present to the future:** if we decide to take a shortcut now, instead of investing time in a proper solution, what problems are we creating for ourselves further down the line?



Technical debt - Some exceptions

- Hackathons
- If you're writing a simple script for a one-off task
- Code competitions
- When developing a proof of concept
- When developing a prototype (as long as you make sure it's indeed a prototype that will be thrown away)
- When you're working with a legacy project that will be deprecated, and it's
- only in maintenance mode for a fixed, short-lived period of time (and again,
- provided this is assured)



Code formatting

- The clean code is not all about formatting
- **PEP8** is the most well-known standard
 - **It helps us:** how we should write our programs, in terms of spacing, naming convention, line length.
- Standardized way of structuring the code, the resulting code will look much more familiar.



Documentation

- Good code is self-explanatory but is also well-documented
- Documenting code is not the same as adding comments to it
- Code comments
- Please delete the commented code



Docstrings

- Docstrings are documentation embedded in the source code
- Represent explanation
- Docstrings are not comments; they are documentation
- Provide documentation for a particular component



Docstrings

- Docstrings are also useful to document design and architecture decisions
- Add a docstring to the most important Python modules, functions, and classes -> how the component fits in the overall architecture.
- Documenting the expected input and output of a function is a good practice
- The docstring is not something separated or isolated from the code. It becomes part of the code it requires manual and constant maintenance.
- They have to be detailed



Some ideas for a clean code

- 1 function = 1 responsibility
- YAGNI principle: “You aren’t gonna need it”
- KISS: “Keep it simple, stupid” or “keep it stupid simple”
- Descriptive Names
- Have fewer than 4 arguments -> Class



Some ideas for a clean code

- Duplication

```
1 t_train = []
2 for file in tweets_train_files:
3     train_doc_1=[r.text for r in ET.parse(join(path_train, file)).getroot()[0]]
4     t_train.append(' '.join(t for t in train_doc_1))
5
6
7 t_test = []
8 for file in tweets_test_files:
9     test_doc_1=[r.text for r in ET.parse(join(path_test, file)).getroot()[0]]
10    t_test.append(' '.join(t for t in test_doc_1))
```

load_data_example.py hosted with ❤ by GitHub

[view raw](#)

Source:

<https://towardsdatascience.com/python-clean-code-6-best-practices-to-make-your-python-functions-more-readable-7ea4c6171d60>

```
1  from typing import List
2
3  def extract_texts_from_multiple_files(path_to_file: str, files: list) -> List[str]:
4
5      all_docs = []
6      for file in files:
7          list_of_text_in_one_file =[r.text for r in ET.parse(join(path_to_file, file_name)).getroot()
8          text_in_one_file_as_string = ' '.join(t for t in list_of_text_in_one_file)
9          all_docs.append(text_in_one_file_as_string)
10
11      return all_docs
```

```
1  from typing import List
2
3  def extract_texts_from_multiple_files(path_to_file: str, files: list) -> List[str]:
4
5      all_docs = []
6      for file in files:
7          text_in_one_file = extract_texts_from_each_file(path_to_file, file)
8          all_docs.append(text_in_one_file)
9
10     return all_docs
11
12 def extract_texts_from_each_file(path_to_file: str, file_name: list) -> str:
13
14     list_of_text_in_one_file =[r.text for r in ET.parse(join(path_to_file, file_name)).getroot()[0]
15     text_in_one_file_as_string = ' '.join(t for t in list_of_text_in_one_file)
16
17     return text_in_one_file_as_string
```



Modal Verb

- Necessity
- Giving advice
- Prohibition



¿What are the modal verbs?

- Modal verbs are used to express certain ,hypothetical conditions, such as advice, capability, or requests
- They're used alongside a main verb to change its meaning slightly.

Subject + modal + verb + complement



¿What are the modal verbs?

- Modal verbs are used to express certain ,hypothetical conditions, such as advice, capability, or requests
- They're used alongside a main verb to change its meaning slightly.



Of giving advice (should - shouldn't)

- Use modals of advice to talk about what **is or isn't a good idea**.
- Modals are followed by the simple form of a verb
 - You should add comments in your code.
 - You should sign up for the hackathon.
 - You shouldn't take that office job.



Of giving advice (Had better)

- **Had better** is stronger than should or ought to. It means something bad could happen if the advice isn't followed
- Use maybe, perhaps or I think with modals to make the advice sound gentler and friendlier.
 - You had better talk with your boss before you buy the plane's tickets.
 - I had better not miss any more days of work.



Of giving advice (Had better)

- Use maybe, perhaps, or I think with modals to make the advice sound gentler and friendlier.
 - Maybe you should become a health care worker



Of necessity

- Use **must + verb** in writing or formal speaking to say that something is necessary or is a rule.
 - Use **have to or need to + verb** in informal speaking.
 - Use **have + to verb** for extra emphasis
-
- Use **don't have to + verb** to say that something is not necessary
 - **Have to and need to** can be used with different verb tenses



Of necessity

- Use **must + verb** in writing or formal speaking to say that something is necessary or is a rule.
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-
- Candidates **must** satisfy the general conditions for admission.
 - You **have to** write the report for tomorrow.
 - We **really need to** learn another language apart from English.



Of prohibition

- **Must not and can't** mean that something is not allowed. There is a law or rule against it.
- This meaning is different from don't have to
 - You must not take pictures here = pictures are not allowed
 - You don't have to take pictures here = pictures are Ok but not necessary



Situations

- In your job
- In a police station
- A friend who will lose the course



Let's practice

https://elt.oup.com/student/solutions/elementary/grammar/grammar_05_022e?cc=pe&sellLanguage=en

<https://test-english.com/grammar-points/a2/should-shouldnt/>

<https://www.englishexercises.org/makeagame/viewgame.asp?id=1734>

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